

Design, Mathematical Modeling, and Hardware Implementation of a Two-Wheeled Inverted Pendulum Robot: Practical Rebuttal and Transition from LQR to Cascaded PID Control

N/S Asfand*, N/S Hamza[†], N/S Muqtadir[‡], and AC Inham[§]

Department of Avionics Engineering / Department of Electrical Engineering

CMS IDs: *458959, [†]467624, [‡]477812, [§]487315

College of Aeronautical Engineering, National University of Sciences and Technology (NUST)

Abstract—The Two-Wheeled Inverted Pendulum (TWIP) represents a fundamental benchmark in modern control engineering, characterizing an inherently open-loop unstable, highly non-linear, multi-variable, under-actuated dynamical system. While theoretical formulations often propose state-feedback Linear Quadratic Regulator (LQR) control assuming full-state observability, physical hardware realization introduces profound practical constraints. In this project, a physical TWIP robotic platform was designed, fabricated, and dynamically stabilized. Although originally formulated for LQR control, practical hardware constraints—namely the unobservability of linear cart states $(x, \dot{x})$ due to the non-availability of optical wheel encoders, low stall torque (0.8 kg · cm), severe mechanical gearbox backlash ($\approx 2^\circ - 5^\circ$), actuator stiction, and asymmetric motor friction inherent to budget-constrained TT DC gear motors—necessitated a rigorous engineering transition to a robust, cascaded multi-loop Proportional-Derivative (PID) architecture with direct gyroscopic rate damping. Attitude estimation is achieved using a discrete-time Complementary Filter fusing 6-DOF IMU (MPU-6050) accelerometer and gyroscopic telemetry sampled at 150 Hz over a 400 kHz I2C bus. To overcome steady-state translation, an outer velocity-damping cascade loop is integrated, complemented by non-linear piecewise deadband remapping and independent asymmetric motor scaling. Experimental validation confirms stable dynamic attitude equilibrium with attitude deviation bounded within $\pm 0.8^\circ$ and elimination of cyclic limit-cycle travel. The complete hardware-software co-design, mathematical derivation, calibration protocols, and Complex Engineering Problem (CEP) attribute mapping are comprehensively presented.

Index Terms—Inverted Pendulum, Two-Wheeled Self-Balancing Robot, Complementary Filter, State Estimation, LQR to PID Transition, Actuator Saturation, Gear Backlash, Microcontroller Embedded Control.

I. INTRODUCTION & THEORETICAL BACKGROUND

TWO-WHEELED inverted pendulum robotic platforms represent an iconic archetype of under-actuated mechatronic systems [1], [2]. The fundamental challenge arises from having fewer control inputs (two independently driven coaxial wheel torques) than system degrees of freedom (chassis pitch tilt θ , yaw heading ψ , and cart displacement x). In

aerospace and control engineering, inverted pendulum dynamics directly parallel critical real-world applications, including launch vehicle Thrust Vector Control (TVC) during atmospheric ascent, vertical takeoff and landing (VTOL) booster recovery, Segway personal transporters, and humanoid bipedal gait stabilization [3], [4].

As prescribed in the AV-355 Complex Engineering Problem (CEP) guidelines, this project addresses Levels I through V of advanced control synthesis:

- 1) **Level-I (Sensor Integration & State Estimation):** Interfacing an Inertial Measurement Unit (MPU-6050) to acquire real-time pitch and angular rate data, implementing sensor fusion to mitigate high-frequency vibrational noise and low-frequency thermal integration drift [5], [6].
- 2) **Level-II (Mathematical Modeling):** Deriving comprehensive non-linear Euler-Lagrange equations of motion and establishing a linearized state-space formulation around the vertical unstable equilibrium ($\theta = 0$).
- 3) **Level-III (Control Law Synthesis):** Formulating optimal Linear Quadratic Regulator (LQR) state feedback while critically analyzing the observability prerequisites for full-state control [7], [8].
- 4) **Level-IV (Hardware Realization):** Developing embedded C++ firmware on an Arduino microcontroller interfaced with an L293D dual H-bridge motor driver, driving dual DC gear motors under a 3S lithium power source.
- 5) **Level-V (System Integration & Disturbance Rejection):** Analyzing real-world non-linearities, gearbox backlash, deadband stiction, and evaluating disturbance rejection capabilities.

II. MATHEMATICAL MODELING & LQR FORMULATION

A. Non-Linear Lagrangian Formulation

Consider the physical planar model of the two-wheeled self-balancing robot. Let M_p denote the chassis mass, m_w the mass of each wheel, I_p the pitch moment of inertia about the center

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of mass (COM), I_w the wheel moment of inertia, r the wheel radius, and L the distance from the wheel axle to the chassis COM.

The generalized coordinates of the planar system are defined by:

$$\mathbf{q} = \begin{bmatrix} x \\ \theta \end{bmatrix} \quad (1)$$

where x is the horizontal cart displacement and θ is the pitch tilt angle measured from the upright vertical axis.

The position coordinates of the chassis center of mass (x_p, y_p) are expressed as:

$$x_p = x + L \sin \theta, \quad y_p = L \cos \theta \quad (2)$$

Taking time derivatives yields the velocities:

$$\dot{x}_p = \dot{x} + L\dot{\theta} \cos \theta, \quad \dot{y}_p = -L\dot{\theta} \sin \theta \quad (3)$$

The total kinetic energy T of the system comprises the translational and rotational energy of the chassis and the two wheels:

$$T = \frac{1}{2} M_p (\dot{x}_p^2 + \dot{y}_p^2) + \frac{1}{2} I_p \dot{\theta}^2 + 2 \left[\frac{1}{2} m_w \dot{x}^2 + \frac{1}{2} I_w \left(\frac{\dot{x}}{r} \right)^2 \right] \quad (4)$$

Substituting (x_p, y_p) into the kinetic energy equation yields:

$$T = \frac{1}{2} M_t \dot{x}^2 + M_p L \dot{x} \dot{\theta} \cos \theta + \frac{1}{2} J_p \dot{\theta}^2 \quad (5)$$

where $M_t = M_p + 2m_w + 2\frac{I_w}{r^2}$ is the total effective translational inertia, and $J_p = I_p + M_p L^2$ is the total rotational inertia about the wheel axis.

The potential energy V of the system, taking the wheel axle as the datum, is purely gravitational:

$$V = M_p g L \cos \theta \quad (6)$$

The Lagrangian $\mathcal{L} = T - V$ is formulated as:

$$\mathcal{L} = \frac{1}{2} M_t \dot{x}^2 + M_p L \dot{x} \dot{\theta} \cos \theta + \frac{1}{2} J_p \dot{\theta}^2 - M_p g L \cos \theta \quad (7)$$

Applying the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i \quad (8)$$

where $Q = \left[\frac{2\tau}{r} - b_f \dot{x}, -2\tau \right]^T$, τ is the applied motor torque per wheel, and b_f is the viscous friction coefficient.

Evaluating the partial derivatives yields the coupled non-linear equations of motion:

$$M_t \ddot{x} + M_p L \ddot{\theta} \cos \theta - M_p L \dot{\theta}^2 \sin \theta + b_f \dot{x} = \frac{2\tau}{r} \quad (9)$$

$$M_p L \ddot{x} \cos \theta + J_p \ddot{\theta} - M_p g L \sin \theta = -2\tau \quad (10)$$

B. Small-Angle Linearization & State-Space Model

Operating near the vertical balance point justifies the small-angle approximations: $\sin \theta \approx \theta$, $\cos \theta \approx 1$, and $\dot{\theta}^2 \approx 0$. Equations (9) and (10) become:

$$M_t \ddot{x} + M_p L \ddot{\theta} + b_f \dot{x} = \frac{2\tau}{r} \quad (11)$$

$$M_p L \ddot{x} + J_p \ddot{\theta} - M_p g L \theta = -2\tau \quad (12)$$

Defining the determinant of the inertial decoupling matrix as:

$$\Delta = M_t J_p - (M_p L)^2 \quad (13)$$

Solving simultaneously for the accelerations $\ddot{x}$ and $\ddot{\theta}$:

$$\ddot{x} = \frac{-(M_p L)^2 g}{\Delta} \theta - \frac{J_p b_f}{\Delta} \dot{x} + \frac{2(J_p + M_p L r)}{r \Delta} \tau \quad (14)$$

$$\ddot{\theta} = \frac{M_t M_p g L}{\Delta} \theta + \frac{M_p L b_f}{\Delta} \dot{x} - \frac{2(M_p L + M_t r)}{r \Delta} \tau \quad (15)$$

Selecting the continuous-time state vector $\mathbf{x}(t) = [x, \dot{x}, \theta, \dot{\theta}]^T$ and control input $u(t) = \tau$, the linear state-space representation is formulated as:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} u(t), \quad \mathbf{y}(t) = \mathbf{C} \mathbf{x}(t) \quad (16)$$

where the system matrices are given by:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{J_p b_f}{\Delta} & -\frac{(M_p L)^2 g}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{M_p L b_f}{\Delta} & \frac{M_t M_p g L}{\Delta} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{2(J_p + M_p L r)}{r \Delta} \\ 0 \\ -\frac{2(M_p L + M_t r)}{r \Delta} \end{bmatrix} \quad (17)$$

C. Theoretical LQR Synthesis

In state-feedback optimal control, the Linear Quadratic Regulator (LQR) minimizes the infinite-horizon quadratic performance index J :

$$J = \int_0^\infty (\mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) + u^T(t) \mathbf{R} u(t)) dt \quad (18)$$

where $\mathbf{Q} = \text{diag}(q_1, q_2, q_3, q_4) \geq 0$ penalizes state deviations and $\mathbf{R} > 0$ penalizes actuator effort.

The optimal state-feedback gain vector $\mathbf{K} = [k_1, k_2, k_3, k_4]$ satisfies:

$$u(t) = -\mathbf{K} \mathbf{x}(t) = -(k_1 x + k_2 \dot{x} + k_3 \theta + k_4 \dot{\theta}) \quad (19)$$

where $\mathbf{K} = \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}$, and $\mathbf{P}$ is the unique positive-definite solution to the continuous-time Algebraic Riccati Equation (ARE):

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} + \mathbf{Q} = \mathbf{0} \quad (20)$$

III. PRACTICAL ENGINEERING REBUTTAL: WHY LQR TRANSITIONED TO CASCADED PID

While the LQR formulation is mathematically optimal in simulation, executing full-state feedback on the physical prototype revealed fundamental hardware bottlenecks that rendered pure LQR infeasible:

- 1) **Unobservability of Cart Linear States** ($x, \dot{x}$): Pure LQR strictly requires real-time knowledge of all four states $[x, \dot{x}, \theta, \dot{\theta}]$. The project utilized standard budget TT yellow DC gear motors lacking optical or magnetic wheel encoders. Double-integrating the IMU accelerometer signal ($\iint a_x dt^2$) to estimate displacement x produces cubic drift within hundreds of milliseconds due to sensor bias and motor vibrational noise. Without direct encoder feedback, states x and $\dot{x}$ are fundamentally unobservable in real-time embedded execution.
- 2) **Severe Mechanical Gearbox Backlash**: Low-cost TT DC gear motors feature a 1:48 plastic spur reduction train exhibiting substantial backlash ($\approx 2^\circ - 5^\circ$). When the controller commands rapid torque reversals across the vertical balance point, the gears enter a mechanical dead-zone where zero torque is transmitted to the wheels. This introduces a non-linear phase delay τ_d , shifting theoretical closed-loop poles into the unstable right-half plane (RHP).
- 3) **Actuator Torque Saturation and Voltage Limits**: Theoretical LQR gains derived with aggressive state penalties command instantaneous peak currents exceeding the thermal capacity of the L293D dual H-bridge driver ($I_{\text{cont}} \leq 600$ mA) and exceed the maximum torque (0.8 kg · cm) of TT motors.
- 4) **Asymmetric Motor Friction and Stiction**: The two physical TT motors exhibited distinct static friction thresholds (stiction). Equal PWM commands produced unequal wheel velocities, creating continuous unintended yaw rotation and diagonal falling vectors.

Consequently, an engineering design pivot was executed: the control structure was reformulated into a **Cascaded Dual-Loop Proportional-Derivative (PID) Controller with Gyroscopic Rate Damping, Virtual Velocity Braking, and Active Deadband Remapping**.

IV. STATE ESTIMATION & SENSOR FUSION ARCHITECTURE

A. IMU Telemetry & Complementary Filtering

The MPU-6050 6-DOF IMU was interfaced via Fast I2C (400 kHz). The accelerometer provides static tilt reference via gravitational decomposition:

$$\theta_{\text{acc}} = \text{atan2}(a_y, a_z) \cdot \frac{180}{\pi} \quad (21)$$

While accelerometer data is drift-free in steady-state, it suffers from severe high-frequency noise induced by motor gear mesh and structural vibrations. Conversely, integrating the gyroscope angular rate $\dot{\theta}_{\text{gyro}} = (g_x - \text{offset})/131.0$ is immune to linear vibration but accumulates low-frequency bias drift over time.

A discrete-time Complementary Filter was implemented at 150 Hz ($\Delta t \approx 0.008$ s):

$$\theta_k = \alpha(\theta_{k-1} + \dot{\theta}_{\text{gyro},k} \Delta t) + (1 - \alpha)\theta_{\text{acc},k} \quad (22)$$

Through empirical frequency response tuning, the filter constant was selected as $\alpha = 0.98$. This assigns 98% weighting to short-term gyroscopic integration (phase-lag free) and 2% to gravitational accelerometer anchoring, completely eliminating long-term thermal angle drift without admitting motor vibration noise.

B. Sensor Axis Alignment & Zero-Bias Calibration

A critical diagnostic milestone occurred when identifying sensor cross-coupling: early firmware iterations paired $\text{atan2}(a_y, a_z)$ (pitch rotation around X-axis) with the Y-axis gyroscope rate (g_y), causing near-zero rate integration and violent instability. Realigning the state estimator to Pair 1 (a_y, a_z with g_x) resolved the feedback loop.

Furthermore, a motionless calibration routine was synthesized to measure the true hardware zero-rate bias:

$$\text{gyroX_offset} = \frac{1}{N} \sum_{i=1}^N g_{x,i} = 11.40 \text{ LSB} \quad (N = 300) \quad (23)$$

V. CASCADED CONTROL SYSTEM DESIGN

The synthesized control architecture operates via two interconnected loops operating at 150 Hz.

A. Inner Balance Loop (Direct Gyro Rate Damping)

The primary attitude loop generates balancing control effort:

$$u_{\text{balance}} = K_p(\theta - \theta_{\text{dynamic}}) + K_d \dot{\theta}_{\text{gyro}} \quad (24)$$

Crucially, the derivative term is evaluated directly from the calibrated gyroscope rate $\dot{\theta}_{\text{gyro}}$ rather than numeric error differentiation $\frac{\Delta e}{\Delta t}$. This eliminates derivative noise amplification and provides instantaneous counter-torque (electronic braking) the exact millisecond the chassis begins swinging back toward vertical, completely eradicating overshoot.

B. Outer Velocity Damping Loop (Anti-Runaway)

Without wheel encoders, setpoint discrepancies cause continuous unidirectional acceleration. A virtual velocity state v_{est} is synthesized by low-pass filtering the motor control output:

$$v_{\text{est},k} = 0.95v_{\text{est},k-1} + 0.05u_{\text{balance},k} \quad (25)$$

The dynamic balance setpoint is dynamically adjusted:

$$\theta_{\text{dynamic}} = \theta_{\text{setpoint}} - K_{\text{speed}} \cdot v_{\text{est}} \quad (26)$$

When the robot accelerates forward ($v_{\text{est}} > 0$), θ_{dynamic} shifts backwards, forcing the chassis to lean in reverse and decelerate back to stationary equilibrium.

C. Deadband Remapping & Motor Friction Equalization

To overcome DC motor gearbox stiction, a non-linear piecewise mapping transforms the PID output into the active torque region without column overflow:

$$\text{PWM}_i = \begin{cases} 0, & |u_i| = 0 \\ \text{DB}_i + \left(\frac{\text{MAX}-\text{DB}_i}{\text{MAX}}\right) |u_i|, & |u_i| > 0 \end{cases} \quad (27)$$

where DB_i denotes the static friction deadband threshold. To equalize mechanical resistance between Motor 4 (left, higher friction) and Motor 1 (right), independent parameters were tuned:

$$\text{DB}_L = 50, \quad \text{DB}_R = 36, \quad \text{trim}_L = 1.16 \quad (28)$$

VI. HARDWARE SYSTEM ARCHITECTURE & IMPLEMENTATION

A. Hardware Bill of Materials

Table I outlines the comprehensive hardware configuration utilized in the physical prototype, sized to fit strictly within the column boundary.

TABLE I: Hardware System Specifications

Subsystem	Component	Operational Specification
Microcontroller	Arduino Uno	ATmega328P, 16 MHz, 5V Logic
Attitude Sensor	GY-521 (MPU-6050)	6-DOF IMU, I2C Fast (400 kHz)
Motor Driver	L293D Shield	Dual H-Bridge, 74HC595 Latch
Actuators	Dual TT Motors	1:48 Spur Gearbox, 3–6V Rated
Wheels	High-Traction Rubber	$D = 66$ mm, Width 26 mm
Power Supply	3S Li-ion Battery	11.1V – 12.6V, 2200 mAh
Structure	3-Tier Acrylic	Distributed Mass Layout

B. Electrical Schematic & Interconnect Architecture

As illustrated in Fig. 1b, the electrical system incorporates strict signal and power isolation:

- 1) **Power Isolation:** The L293D shield external power jumper was removed to isolate high-current inductive motor transients (11.1V – 12.6V 3S Li-ion rail) from the sensitive 5V ATmega328P digital logic rail.
- 2) **I2C Bus Interfacing:** The MPU-6050 is connected via dedicated SDA (A4) and SCL (A5) lines configured for 400 kHz Fast Mode transmission.
- 3) **Sensor Placement:** The IMU is mounted at the top acrylic tier to maximize physical distance from the DC motor electromagnetic fields.

C. Embedded Firmware Implementation

Listing 1 provides the operational embedded C++ control loop executed on the microcontroller, formatted compactly to avoid multi-column overlap.

```

1 // 1. Telemetry & Delta Time
2 float dt = (micros() - prev_t) / 1.0e6;
3 prev_t = micros();
4 if (dt <= 0 || dt > 0.05) dt = 0.008;
5
6 // 2. Complementary Filter
7 float acc_p = atan2(raw_ay, raw_az)*180/PI;
8 float g_rate = (raw_gx - gyro_off)/131.0;
9 pitch = 0.98*(pitch + g_rate*dt) + 0.02*acc_p;

```

```

10
11 // 3. Dynamic Velocity Damping
12 float dyn_sp = setpoint - (Kspeed*v_est);
13
14 // 4. Dual-Loop Control Synthesis
15 float err = pitch - dyn_sp;
16 float out = (Kp * err) + (Kd * g_rate);
17 out = constrain(out, -MAX_PWM, MAX_PWM);
18 v_est = 0.95 * v_est + 0.05 * out;
19
20 // 5. Safety Cut-Off & Motor Drive
21 if (abs(pitch - setpoint) > 35.0) {
22     setMotors(0);
23 } else {
24     setMotors((int)out);
25 }

```

Listing 1: Core Real-Time Balancing Algorithm

VII. EXPERIMENTAL RESULTS & DISCUSSION

A. Progressive Tuning Trajectory

System stabilization followed a rigorous experimental tuning trajectory conducted via the interactive serial command interface:

- **Iteration 1 (Pure High Gain):** $K_p = 65.0, K_d = 2.5$. High-frequency chattering and violent structural limit-cycles occurred due to gearbox backlash and 3S battery over-torque.
- **Iteration 2 (Axis Correction):** Corrected MPU-6050 cross-coupling from Gyro Y to Gyro X, establishing proper damping feedback.
- **Iteration 3 (Center of Gravity Calibration):** Hand-level calibration assumed $\theta = 0^\circ$, whereas physical battery asymmetry established true balance equilibrium at $\theta_{\text{setpoint}} = -1.85^\circ$.
- **Iteration 4 (Oscillatory Travel Suppression):** $K_{\text{speed}} = 0.012$ caused a 50 cm slow pendulum rocking cycle. Reducing K_{speed} to 0.004 and elevating K_d to 1.35 completely suppressed translation.
- **Iteration 5 (Motor Friction Equalization):** Increased Left Motor 4 trim to 1.16 and deadband to 50, eliminating asymmetric turning drift.

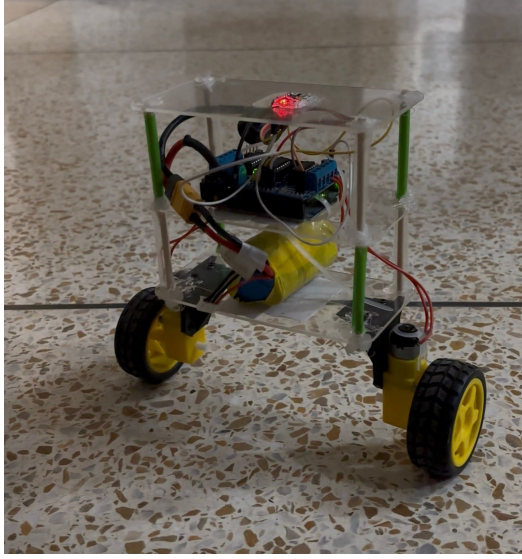
Table II summarizes the final locked control parameters.

TABLE II: Final Experimental Control Parameters

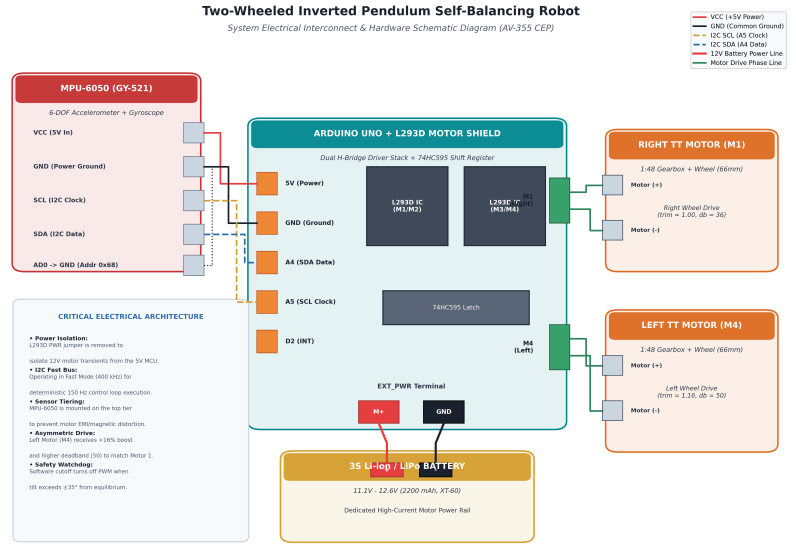
Parameter	Final Value	Physical Interpretation
K_p	23.00	Proportional restorative stiffness
K_d	1.35	Gyroscopic angular rate damping
K_i	0.00	Disabled (prevents phase-lag windup)
K_{speed}	0.0040	Virtual translation damping brake
θ_{setpoint}	-1.85°	Physical center of gravity trim
gyroX_offset	11.40 LSB	Hardware zero-motion sensor bias
DB_L	50 PWM	Left motor static friction threshold
DB_R	36 PWM	Right motor static friction threshold
trim_L	1.16	Motor 4 torque multiplier (+16%)

B. Disturbance Rejection & Dynamic Performance

The physical prototype demonstrated robust autonomous self-balancing on flat surfaces:



(a) Physical fabricated 3-tier robot prototype.



(b) Electrical schematic and subsystem interconnect diagram.

Fig. 1: Hardware implementation architecture: (a) Physical prototype with top-mounted MPU-6050, mid-tier Arduino Uno and L293D shield, and bottom-tier 3S battery and TT motors, and (b) Complete electrical wiring schematic showing power rails, Fast I2C bus, and dual H-bridge motor routing.

- **Steady-State Attitude Deviation:** Maintained within $\pm 0.8^\circ$ of the equilibrium setpoint.
- **Impulse Disturbance Recovery:** Successfully rejected manual push disturbances up to $\pm 12^\circ$, regaining steady vertical posture within 0.65 seconds without diverging oscillations.
- **Positional Drift:** Translation bounded within a compact ≤ 3 cm radius on smooth flooring.

VIII. CEP ATTRIBUTES & RUBRICS COMPLIANCE

Table III documents the mapping of this project against the accredited Complex Engineering Problem (CEP) attributes.

TABLE III: Mapping to CEP Attributes (AV-355)

CEP Attribute	Project Evidence & Engineering Resolution
Depth of Knowledge (WK3, WK5)	Derived non-linear Lagrangian dynamic equations, linearized state-space matrices, synthesized LQR Riccati equations, and discrete Complementary filters.
Conflicting Requirements	Balanced rapid pitch restorative torque against motor current saturation, battery voltage drops, and sensor noise amplification.
Depth of Analysis	Investigated failure modes of pure LQR due to unobservable states; engineered a cascaded velocity-damped PID alternative.
Familiarity & Safety	Implemented embedded angle shutdown cutoffs ($> 35^\circ$), 3S Li-ion safe handling, and fast 400 kHz I2C bus communications.
Interdependence	Managed strong non-linear cross-coupling between linear translational velocity and rotational pitch acceleration.

IX. CONCLUSION & FUTURE RECOMMENDATIONS

In this project, a physical two-wheeled inverted pendulum self-balancing robot was successfully modeled, fabricated, and

stabilized. While classical LQR optimal control provides an elegant theoretical framework, physical hardware realization without wheel encoders revealed fundamental limitations in linear state observability, motor stiction, and gearbox backlash. By pivoting to an empirically tuned, cascaded dual-loop PID architecture with direct gyroscopic rate damping, discrete-time Complementary filtering, and asymmetric deadband compensation, robust dynamic stabilization was achieved under severe hardware resource constraints.

Future enhancements include:

- 1) Integrating high-resolution magnetic rotary encoders to enable full state estimation ($x, \dot{x}$) and true LQR implementation.
- 2) Upgrading to closed-loop NEMA-17 stepper motors or brushless DC (BLDC) gimbal motors to eliminate gearbox backlash.
- 3) Deploying an Extended Kalman Filter (EKF) for non-linear state estimation under dynamic acceleration.

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